Instant Particle Size Distribution Measurement Using CNNs Trained on Synthetic Data

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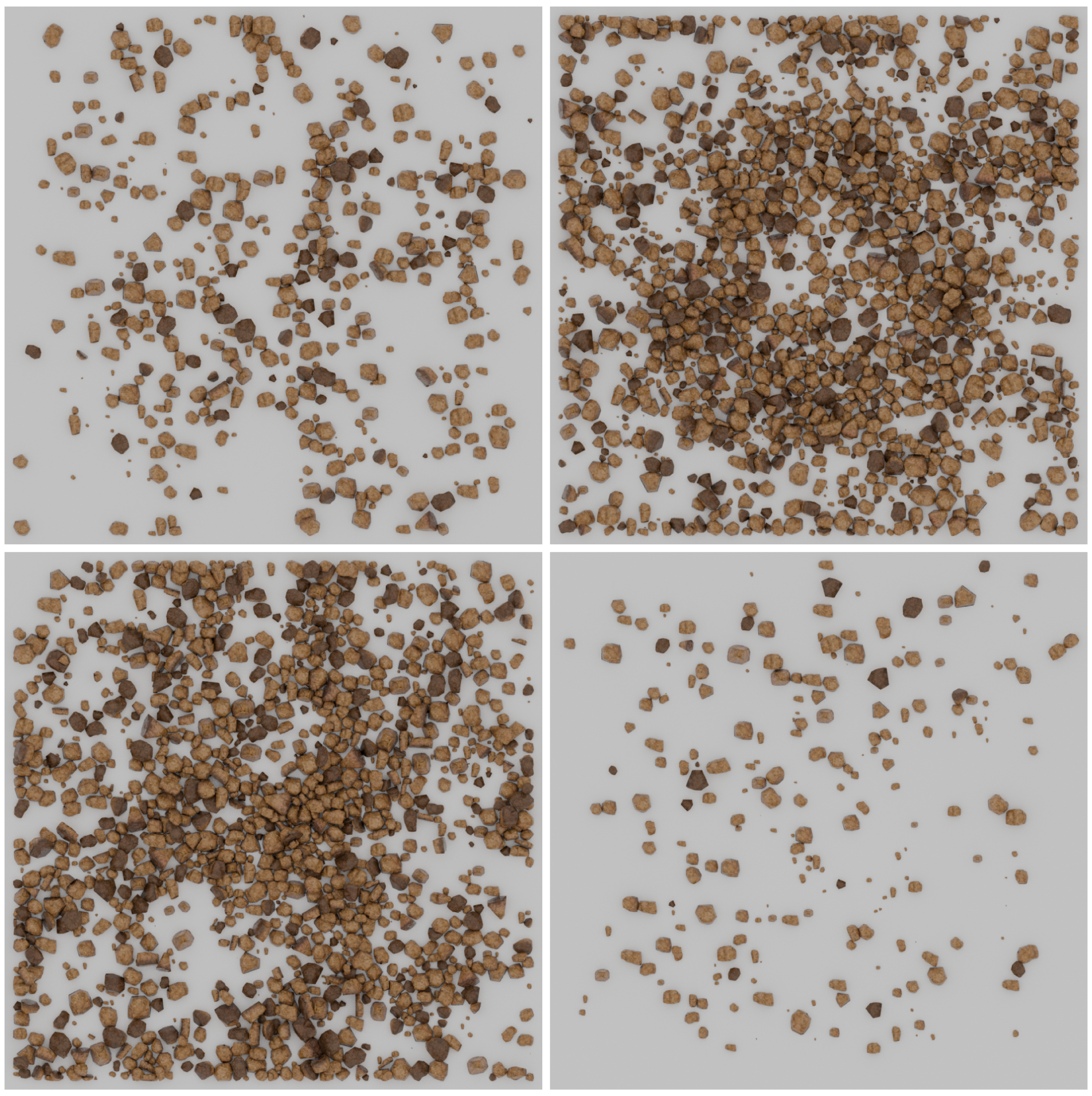
Motivation

Context: Particle size distribution is a key characteristic that influences how phosphate behaves in various contexts, including phosphate rock processing, soil chemistry, and industrial applications. The distribution, or the proportion of particles in a given size range, directly affects important properties like surface area, reactivity, and physical handling.

Problem: Traditional methods are costly (data collection, labeling, etc.) and slow. Creating a high-quality, diverse dataset with reliable labels is time-consuming, error-prone, and difficult to scale. It is a real pain to make such a dataset manually.

Our Solution: We leverage realistically rendered synthetic data generated with Blender to provide automatic, perfectly labeled supervision for key PSD metrics (d10, d50, d90), enabling robust CNN training without manual annotation. We open-source our data generation pipeline and training framework to accelerate research in automated PSD estimation.



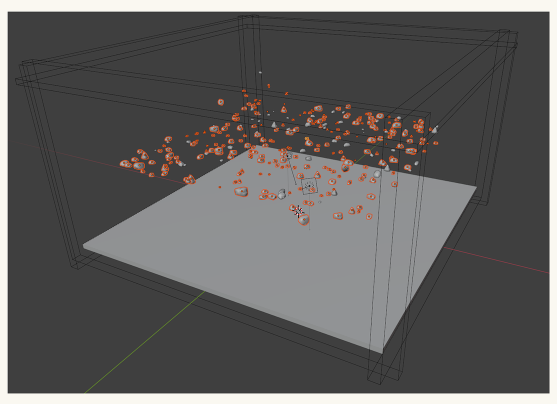
Realistic synthetic particle images with controlled size distributions and perfect labels.

Methodology

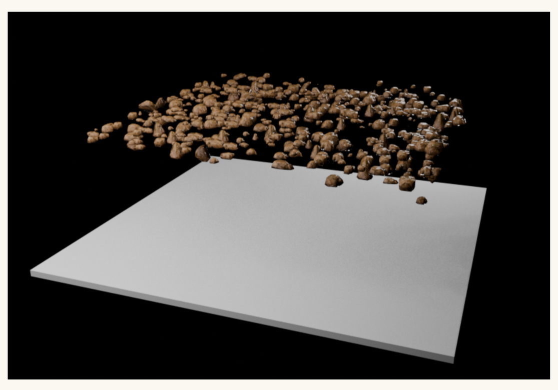
Synthetic Data Generation Pipeline

We developed a fully automated Blender-based pipeline with two main components:

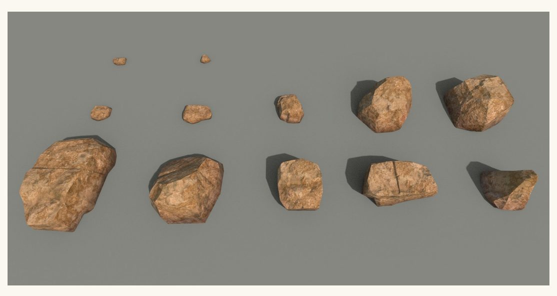
- **Metadata Generation:** Randomly samples particle count (700-1000), sizes from truncated normal distribution $\mathcal{N}_T(\mu, \sigma^2; a, b)$ where $\mu \in [6, 12]$mm, $\sigma \in [6, 8]$mm, truncated to $[a, b] = [0.1, 20]$mm, and positions. Outputs structured JSON with exact PSD ground truth (d10, d50, d90).
- **Scene Construction & Rendering:** Loads 3D particle meshes with realistic textures (base color, metallic, roughness, normal maps), runs rigid-body physics simulation, renders with Cycles engine (GPU-accelerated). 4,500 images rendered on HPC A100 GPUs.



Blender setup



Rendered scene



Particle models

CNN Training

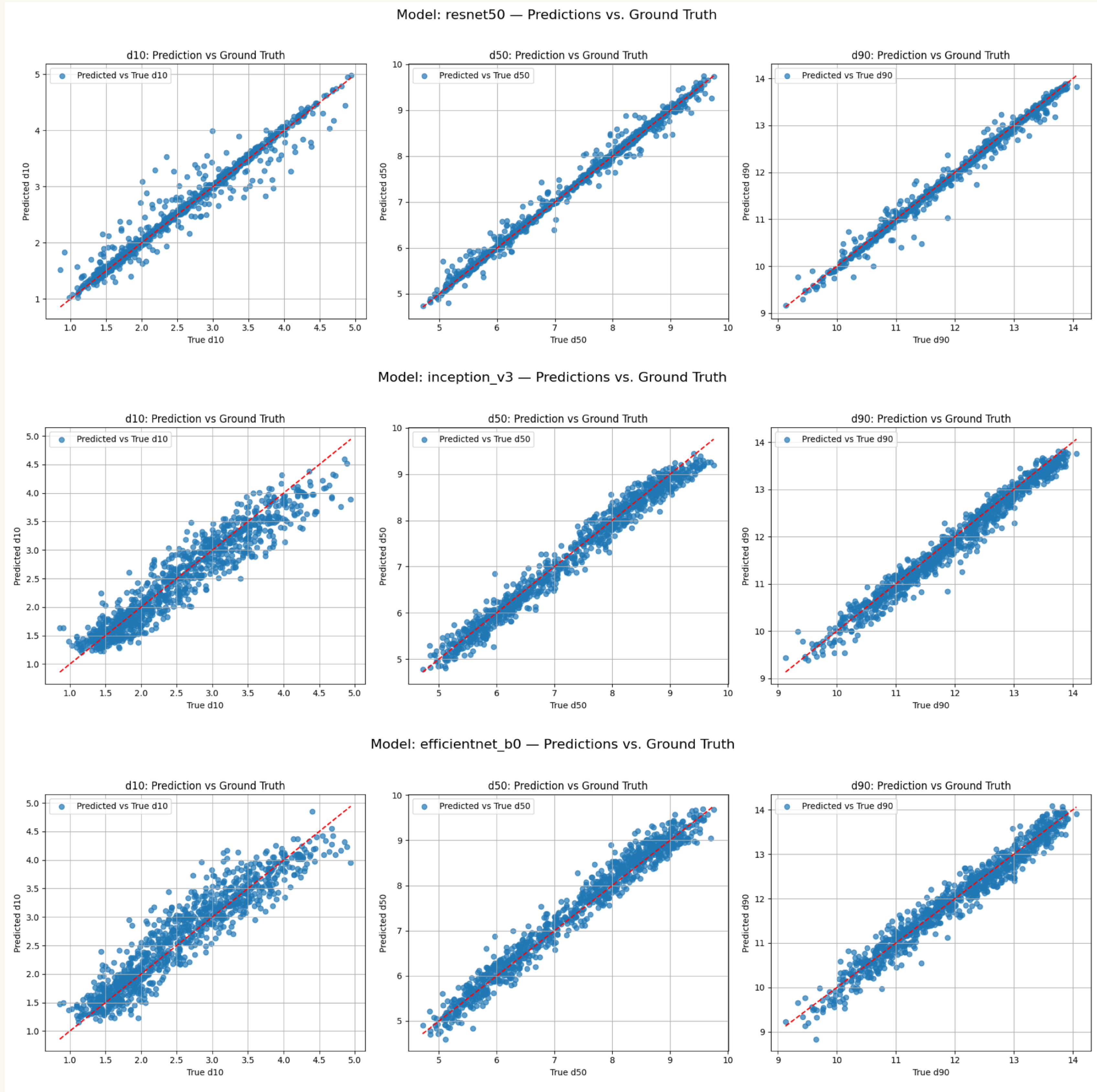
Three architectures evaluated: ResNet-50, InceptionV3, EfficientNet-B0 (all ImageNet-pretrained).

- Fine-tuned for regression (d10, d50, d90) with MSE loss, Adam optimizer (lr=10^{-4})
- Data augmentation: rotations (0°, 90°, 180°, 270°), horizontal flips
- Trained 50 epochs on HPC A100 GPUs (3 hours per model)

Experimental Results

Prediction Accuracy

All three models achieve high accuracy on synthetic test data. ResNet-50 shows marginally better performance (R^2=0.9987, MAE=0.077mm), but differences are minor (<0.1mm) across d10, d50, d90 predictions.



Predicted vs. ground truth PSD values on synthetic test data for all three CNN models.

Computational Efficiency

EfficientNet-B0 offers best speed/accuracy trade-off: 37.75 FPS (GPU) vs. ResNet-50's 27.31 FPS, with 5× fewer parameters (5.3M vs. 25.6M), making it ideal for real-time industrial deployment on commodity hardware.

Table 1. Model performance and efficiency comparison on laptop hardware (RTX 3060 GPU, AMD Ryzen 7 5800H CPU).

CNN Model	R^2 Score	GPU FPS	CPU FPS	Time/Img (s)	Params (M)
ResNet-50	**0.9987**	27.31	14.66	0.0366	25.6
InceptionV3	0.9966	23.18	12.00	0.0431	23.9
EfficientNet-B0	0.9959	**37.75**	**21.51**	**0.0265**	**5.3**

Conclusion & Future Work

Our Contribution: We present an open-source, fully automated Blender-based pipeline for generating realistic synthetic particle datasets with perfect PSD labels (d10, d50, d90). This eliminates costly manual annotation while enabling robust CNN training for real-time industrial PSD measurement.

Results: ResNet-50 achieves highest accuracy (R^2=0.9987), while EfficientNet-B0 delivers optimal speed/accuracy balance (38 FPS, 5.3M params) for deployment. This work was accepted and published at the *Synthetic Data for Computer Vision Workshop, CVPR 2025, Nashville, TN*.

Future Work: Domain adaptation with incremental fine-tuning on limited real data to bridge the synthetic-to-real gap, and edge device deployment for decentralized industrial monitoring.

Open-Source Resources

The paper and the code for data generation and training are made public. You can access them by scanning the QR codes below.

SynData4CV CVPR Paper

Source Code